



OPEN Cross-modal gated feature enhancement for multimodal emotion recognition in conversations

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Emotion recognition in conversations (ERC), which involves identifying the emotional state of each utterance within a dialogue, plays a vital role in developing empathetic artificial intelligence systems. In practical applications, such as video-based recruitment interviews, customer service, health monitoring, intelligent personal assistants, and online education, ERC can facilitate the analysis of emotional cues, improve decision-making processes, and enhance user interaction and satisfaction. Current multimodal emotion recognition research faces several challenges, such as ineffective emotional information extraction from single modalities, underused complementary features, and inter-modal redundancy. To tackle these issues, this paper introduces a cross-modal gated attention mechanism for emotion recognition. The method extracts and fuses visual, textual, and auditory features to enhance accuracy and stability. **A cross-modal guided gating mechanism is designed to strengthen single-modality features and utilize a third modality to improve bimodal feature fusion, boosting multimodal feature representation.** Furthermore, **a cross-modal distillation loss function is proposed to reduce redundancy and improve feature discrimination.** This function employs a **dual-supervision mechanism with teacher and student models**, ensuring consistency in single-modal, bimodal, and trimodal feature representations. Experimental results on the IEMOCAP and MELD datasets indicate that the proposed method achieves higher accuracy and comparable F1 scores than existing approaches, highlighting its effectiveness in capturing multimodal dependencies and balancing modality contributions.

Keywords Deep learning, Cross modal, Emotion recognition in conversations, Transformer, Gated attention

In recent years, the rapid development of artificial intelligence and human-computer interaction technologies has positioned emotion recognition as a critical area of research across a wide range of fields^{1–3}. In human-computer interaction, emotion recognition can significantly enhance the capabilities of intelligent assistants, enabling more personalized, intuitive, and responsive user interactions⁴. In video-based recruitment interviews, it plays a pivotal role in analyzing the correlation between emotional cues and interview outcomes, ultimately improving the accuracy and reliability of hiring decisions^{5,6}. Additionally, emotion recognition technology is also being leveraged in various other applications, such as customer service⁷, health monitoring⁸, intelligent personal assistants⁹, and online education¹⁰, where it helps improve service quality and user satisfaction. These applications highlight the transformative impact of emotion recognition on enhancing user experience and advancing various sectors.

Emotion recognition methods for conversational scenarios can extract emotional features from multimodal data to accurately assess a user's emotional state. Early approaches to emotion recognition primarily focused on extracting features from single-modal data, including audio^{11,12}, text¹³, and visual information¹⁴. Audio-based emotion recognition typically analyzes features such as frequency, pitch, and prosody to determine emotional states. Text-based methods leverage natural language processing techniques to identify emotional cues through sentiment-bearing words, semantic features, and syntactic structures. Visual-based approaches, on the other hand, concentrate on analyzing non-verbal cues such as facial expressions, eye movements, and micro-

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expressions. While single-modal emotion recognition methods can be effective in certain contexts, they often struggle to adapt to the complexities of real-world emotional expression scenarios.

With the advancement of deep learning^{15,16}, models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and Transformers have been widely applied to emotion recognition tasks. This progress has enabled the extraction and fusion of features across different modalities, making multimodal emotion recognition a growing focus of research¹⁷. By integrating emotional information from multiple modalities, it becomes possible to capture a more comprehensive representation of a user's emotional state. For instance, combining audio, text, and visual information from conversational contexts allows for more accurate assessment of a speaker's emotions. However, multimodal emotion recognition also faces challenges, such as effectively leveraging complementary features across modalities and minimizing the interference of redundant information between them.

In summary, current emotion recognition tasks based on artificial intelligence still face several technical challenges. Traditional single-modal approaches, which rely solely on visual, textual, or auditory information, often fail to comprehensively capture emotional features and result in suboptimal recognition performance due to the inherent limitations of individual modalities. On the contrary, multimodal emotion recognition can more accurately capture emotional characteristics by integrating information from diverse modalities, thereby improving recognition accuracy and robustness. However, how to effectively integrate complementary information across modalities, minimize the impact of redundant information on emotion recognition, and enhance the model's ability to represent features from each modality remain critical challenges that require further research in the field of multimodal emotion recognition.

To address the aforementioned challenges, this paper proposes a multimodal emotion recognition method based on a cross-modal gated attention mechanism. This approach leverages deep features from visual, textual, and auditory modalities, achieving feature fusion through a cross-modal guided gated attention mechanism. During the fusion process, the designed gating network enhances the representation of single-modal features while incorporating a third modality to extract valuable information from bimodal fused features, thereby strengthening inter-modal complementarity. In addition, to further improve the model's emotion recognition performance, a cross-modal distillation loss function is introduced. This function employs a dual-supervision mechanism involving teacher and student models to reduce redundant information across modalities and enhance the consistency of feature representations in single-modal, bimodal, and tri-modal contexts.

Related works

Deep-learning-based emotion recognition methods

In conversational scenarios, data modalities typically include spoken dialogue (audio information), textual expressions (text information), facial expressions (visual features), and electroencephalogram (EEG) signals. Emotion recognition tasks in such contexts primarily rely on the first three modalities—audio, text, and visual information—to infer the emotional states of the participants. A brief summary of emotion recognition methods based on audio, text, and image modalities individually is presented below.

Current image-based emotion recognition methods primarily focus on extracting information from facial regions. Specifically, facial regions are first identified in images using face detection algorithms¹⁸, followed by global feature extraction through deep learning networks such as Residual Networks (ResNet)¹⁹ to perform classification tasks. Alternatively, facial key points are detected to construct a facial structure topology for expression recognition. For instance, Yolcu et al.²⁰ proposed a CNN-based system that concentrates on detecting key facial regions, such as eyebrows, mouth, and eyes. Li et al.²¹ developed a CNN architecture specifically designed to address facial occlusion issues, improving the model's performance on datasets like AffectNet. Liu et al.²² employed a combination of multiple small CNN sub-networks for emotion recognition, achieving high classification accuracy on the FER2013 dataset.

Unlike image-based deep learning methods for emotion recognition, speech-based approaches require more extensive data preprocessing. Commonly used preprocessing techniques for feature extraction include prosodic features, spectral features, voice quality features, and phase features. Following feature extraction, temporal models are employed to perform high-dimensional feature mapping and classification tasks. For example, Bhargale et al.²³ utilized Mel-frequency cepstral coefficients (MFCCs) combined with a one-dimensional convolutional neural network to capture long-term dependencies in temporal features and improve emotion recognition accuracy. Khan et al.²⁴ enhanced the system's emotion recognition capabilities by fusing prosodic features with spectral features, such as pitch and MFCCs. Lee et al.²⁵ adopted spectral-based entropy features, like fast Fourier transform (FFT), to capture more nuanced emotional variations.

Text-based emotion recognition methods accomplish the task by extracting features from textual information. Since raw text data cannot be directly used for computation, it is necessary to encode the text into numerical representations. In subsequent processing, textual data shares some similarities with audio data. For instance, Kusal et al.²⁶ employed word embedding techniques such as Word2Vec or GloVe to integrate sentiment-bearing words into emotion classification models, thereby improving classification performance. Kumar et al.²⁷ utilized Transformer models to capture long-range dependencies in text, achieving higher emotion recognition accuracy. Paul et al.²⁸ applied the BERT model to extract contextual features, enabling fine-grained analysis of textual emotions. Rashid et al.²⁹ leveraged Long Short-Term Memory (LSTM) networks to process sequential text information, effectively capturing contextual emotional features.

Multimodal emotion recognition methods in conversational scenarios

Multimodal fusion methods integrate information from different modalities, such as audio, text, and images, to address the limitations of single-modal approaches. However, since feature vectors from different modalities reside in distinct subspaces, they must be projected into a common subspace for effective fusion. For instance,

deep learning methods based on audio and text primarily capture contextual correlations to identify individual emotions, while images inherently contain spatial rather than temporal information. Therefore, multimodal recognizers incorporating image information first utilize spatial feature extraction models, such as convolutional neural networks (CNNs), to transform spatial information into semantic representations. This allows image features to align with audio and text features, enabling multimodal fusion and emotion recognition. In conversational scenarios, multimodal emotion recognition methods focus on exploring both inter-modal and intra-modal information. Based on whether speaker information is utilized, existing methods can be categorized into two types: speaker-independent methods and speaker-dependent methods.

Speaker-independent methods do not consider individual personality traits in conversations but instead focus on capturing contextual information within the dialogue. For example, Abdul-Al et al.³⁰ proposed a data augmentation technique using Gaussian white noise to mitigate overfitting during training. Ying et al.³¹ and Gonçalves et al.³² employed CNN models to extract audio and visual features, respectively, while capturing inter-modal correlations. Akinpelu et al.³³ utilized short-time Fourier transform and convolutional neural networks to extract audio features, combined with Transformer and long short-term memory (LSTM) networks to process text and image data, significantly improving emotion recognition accuracy. HiGRU³⁴ and AGHMN³⁵ employed gated recurrent units (GRUs) to model contextual relationships between words and utterances. MVN³⁶ leveraged a multi-view network to model dependencies at both word and utterance levels in conversations.

Speaker-dependent methods encode speaker information to enhance the model's prior knowledge. For instance, DialogueRNN³⁷ utilizes GRUs to encode speaker information and integrate it with textual features. DialogueGCN³⁸ employs graph convolutional networks to model speaker information and dialogue sequence. HiTrans³⁹ leverages word embeddings from two Transformers to explore dependencies between speaker and textual features. These methods have achieved significant success in text-based emotion recognition tasks. For multimodal emotion recognition, related work further improves feature fusion effectiveness using graphs and Transformers. For example, DialogueTRM⁴⁰ uses Transformers and multi-granularity interaction fusion modules to explore intra- and inter-modal emotional behaviors. MMGCN⁴¹ employs fully connected graphs to capture long-range contextual information across modalities. MM-DFN⁴² introduces a graph-based dynamic fusion module to reduce inter-modal redundancy and enhance complementarity. MMTr⁴³ automatically adjusts the representation effectiveness of different modalities through Transformers. UniMSE⁴⁴ performs modality fusion at syntactic and semantic levels and incorporates inter-modal contrastive learning to distinguish feature differences between samples. SDT⁴⁵ enhances multimodal feature representation using gated networks and inter-feature supervised learning.

Methodology

Overall pipeline

As illustrated in Fig. 1, this paper proposes a feature fusion method based on a cross-modal gated attention mechanism (Gated Multimodal Attention, GMA) for multimodal emotion recognition tasks. The method aims to enhance emotion recognition accuracy by integrating information features from three modalities: visual, textual, and audio. In this architecture, pre-trained models are first utilized to extract features from the visual, textual, and audio modalities separately. These features are then fed into a Transformer-based feature extraction module, where a cross-modal gated attention mechanism dynamically fuses the multimodal information to capture inter-modal dependencies. The gating mechanism regulates the contribution of each modality to the emotion classification. The proposed method primarily consists of four modules: a multimodal feature extraction module, a cross-modal attention module, a gated feature fusion module, and a classification module.

Our approach introduces a cross-modal gated self-attention mechanism to dynamically prioritize each modality's contribution, addressing the issue of cross-modality redundancy. Unlike MMGCN and MMDFN that focus on graph-based fusion and modality interactions, we use self-attention to refine multimodal representations, improving emotion classification. Additionally, we propose a cross-modal distillation loss with a dual-supervision mechanism, ensuring consistent feature representations across single-modal, bimodal, and

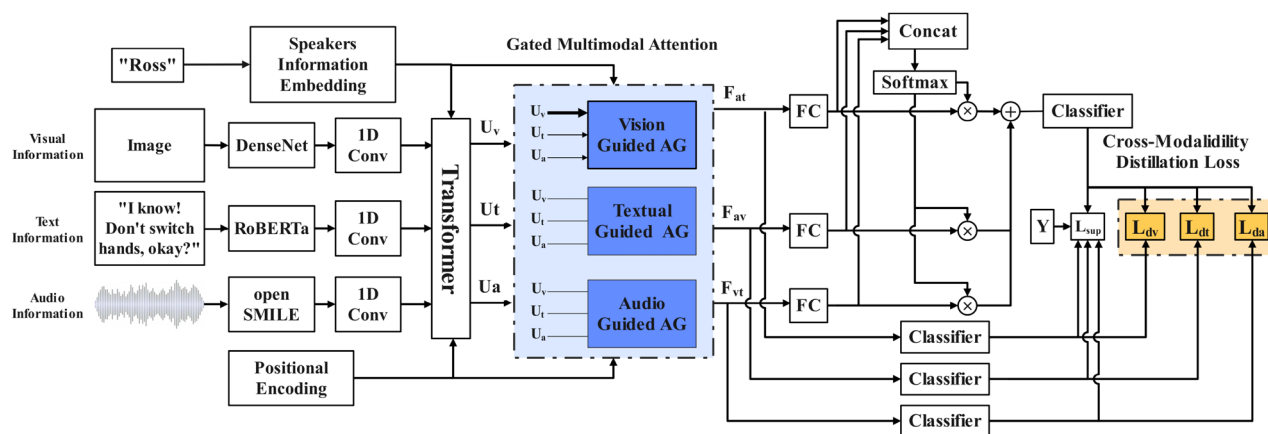


Fig. 1. The pipeline of the proposed method.

trimodal settings. This has improved the coherence of multimodal features, particularly for complex emotion categories. Experimental results show that our method outperforms MMGCN, MMDFN, SDT and a few other models, especially in balancing modality contributions.

To ensure a fair comparison of the data, this study adopts the same feature extraction methods as previous works^{41–43} to process information from three modalities. Specifically, the RoBERTa large model⁴⁶ is utilized to extract textual features, resulting in feature vectors of dimension 1024. Acoustic features are extracted using the openSMILE toolkit⁴⁷, where the feature vector dimensions are 1582 for the IEMOCAP dataset⁴⁸ and 300 for the MELD dataset⁴⁹. For visual feature extraction, a DenseNet model pretrained on the Facial Expression Recognition Plus dataset⁵⁰ is employed, yielding image feature vectors of dimension 342. To align the dimensionality of features across modalities, a one-dimensional convolution (1D Conv) layer is applied to further map the preprocessed features before feeding them into the Transformer model.

The cross-modal attention mechanism proposed in this study facilitates interactions among visual, textual, and acoustic features. By incorporating a gating mechanism, the model dynamically adjusts the contribution weights of different modalities. The fused features are then passed through a fully connected layer and classified using Softmax for emotion recognition. Furthermore, a cross-modal distillation loss function is designed to reduce redundant information across modalities, thereby enhancing the model's performance in emotion recognition.

In the proposed method, speaker ID embeddings are incorporated to encode speaker-specific information. Each speaker ID is mapped to a fixed-dimensional embedding vector through an embedding layer. These embeddings are integrated with the multimodal inputs in two key components.

First, in the Positional Encoding module, speaker embeddings are added to the standard positional encodings and the input features. This design enables the model to jointly consider utterance order and speaker identity during encoding. Second, in each Transformer Encoder layer, the combined representation-including speaker information-is employed to capture dependencies across utterances and maintain awareness of the speaker.

Additionally, the output features containing speaker information are used in the Multimodal Gated Fusion mechanism, in combination with the transformed text, audio, and visual features to produce the final representations used for classification. Through these steps, the model can better distinguish emotional states associated with different speakers and improve recognition accuracy in multi-speaker dialogue contexts.

The architecture of the multimodal transformer

As shown in Fig. 2, this study employs the Encoder component of the Transformer to extract multimodal features. When multimodal data is fed into the Transformer layer, positional encoding is first applied to the three modal features $\{V_v, V_t, V_a\}$. Considering the characteristics of conversational scenarios, speaker identity information is explicitly incorporated into the encoding process. By combining positional encoding with speaker identity embeddings at each time step, the model enhances its understanding of conversational structures and contextual roles. The positional encoding for any given feature x is formulated as follows:

$$x = x + PE + e_s \quad (1)$$

where PE represents the original positional encoding of the Transformer, and e_s is a trainable embedding vector that encodes speaker identity information. Note that x refers to the feature representation to be updated at each

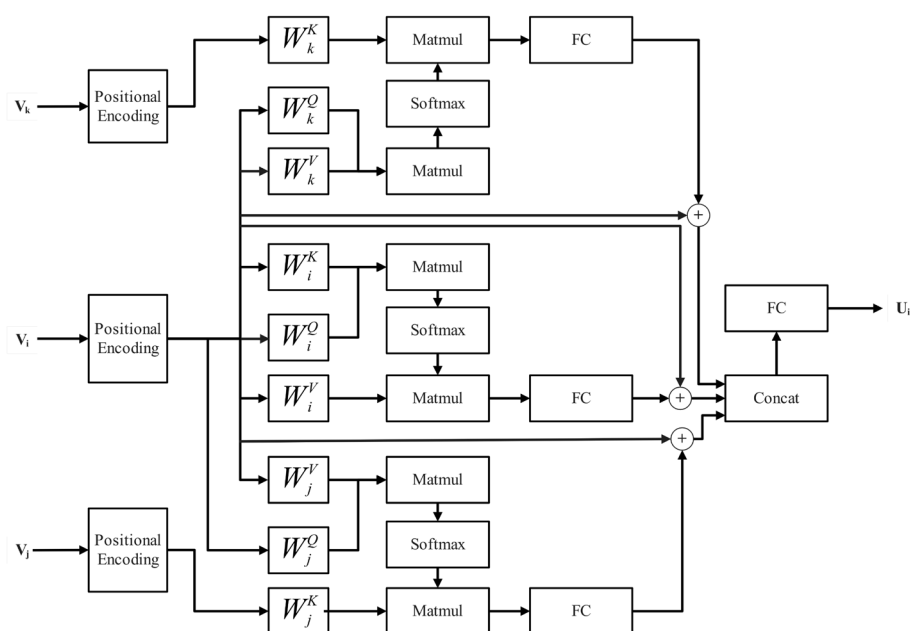


Fig. 2. The architecture of the applied multimodal transformer.

time step. After the addition, x contains both positional and speaker information. The mathematical expression of PE is as follows:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \quad (2)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d}}}\right) \quad (3)$$

where pos denotes the position in the sequence, i represents the dimension index, and d is the encoding dimension. Here, $pos \in 0, 1, \dots, L - 1$ with L being the maximum sequence length.

After incorporating positional encoding information, the features are processed using the multi-head self-attention mechanism. The core principle of multi-head self-attention is to capture different contextual information from the input features through multiple parallel attention heads. Each attention head computes attention scores based on Query (Q), Key (K), and Value (V) and subsequently generates context vectors.

For any given modality feature V_i , the Transformer architecture is divided into modality-specific attention enhancement and cross-modal attention enhancement. In modality-specific attention enhancement, the computation of KQV is performed using the same modality feature V_i as input. The expression for the output feature V_{ii} is formulated as follows:

$$V_{ii} = \text{Softmax}\left(\frac{W_i^Q V_i^T (W_i^K V_i^T)^T}{\sqrt{d_k}}\right) W_i^V V_i^T + V_i \quad (4)$$

where d_k is the dimensionality of the key vectors, used to scale the dot product and prevent excessively large values that could lead to vanishing or exploding gradients. Here, $W_i^Q \in \mathbb{R}^{d_q \times d_h}$, $W_i^K \in \mathbb{R}^{d_k \times d_h}$, and $W_i^V \in \mathbb{R}^{d_v \times d_h}$ are trainable projection weight matrices for computing query, key, and value representations, respectively, where d_h is the hidden dimension of the input features and i refers to a modality from text, audio or visual. Additionally, residual connections are incorporated into the attention mechanism to prevent information loss.

For cross-modal attention enhancement, the overall computation follows the same approach as Eq. (4). However, in the computation of the key vectors, features from other modalities are used as input to capture semantic associations across modalities. The final output features are obtained through concatenation-based fusion:

$$U_i = FC(\text{concat}(V_{ii}, V_{ij}, V_{ik})) \quad (5)$$

Here, V_{ii} , V_{ij} , and V_{ik} refer to attention-enhanced representations of the i^{th} modality interacting with itself and the other two modalities (i.e., text, audio, visual). The operator $FC(\cdot)$ denotes a fully connected transformation applied to the concatenated features to produce the fused representation U_i .

Cross-modal gated attention feature fusion method

To further enhance the correlation among multimodal features in the dialogue process, this paper proposes a cross-modal gated attention feature fusion module. In this module, multiple modality features are dynamically integrated through the attention mechanism, while a gating unit regulates the contribution of each modality to the final feature representation.

As seen in Fig. 3, the input features U_i , U_j and U_k represent features from different modalities. These features are first processed through a fully connected (FC) layer and then passed through a Sigmoid activation function to generate gating coefficients. The gating weights are formulated as follows:

$$G_m = \sigma(W_m U_m + b_m) \quad m \in \{j, k\} \quad (6)$$

where σ represents the Sigmoid function, W_m is a learnable weight matrix, and b_m is the bias term. Here, U_m represents the input features from the m th modality (e.g., text, audio, visual) after the attention-enhanced process. The gating coefficients G_m are generated for each modality using the corresponding weight matrix W_m and bias term b_m . These coefficients are obtained by applying the Sigmoid activation function $\sigma(\cdot)$ to control the information flow across modalities. The weight matrix W_m is a trainable parameter for the m th modality, and b_m is the corresponding bias term.

Unlike previous works, the proposed gated attention mechanism first adjusts the gating features using a single modality and then constructs bimodal fusion features through an additive process, with the third modality providing attention-based guidance. The additive mechanism ensures that bimodal features are extracted simultaneously without increasing the number of model parameters, thereby avoiding the gradient reduction issue caused by multiplicative weight interactions. Furthermore, employing cross-modal attention for bimodal feature extraction enables a deeper exploration of inter-modal mutual information, thereby improving the model's recognition accuracy. Since this study adopts a three-modal feature extraction approach, the final decision-making process involves three sets of fused features, including vision-guided audio-text fusion features F_{at} , test-guided audio-vision fusion features F_{av} , and audio-guided vision-text fusion features F_{vt} .

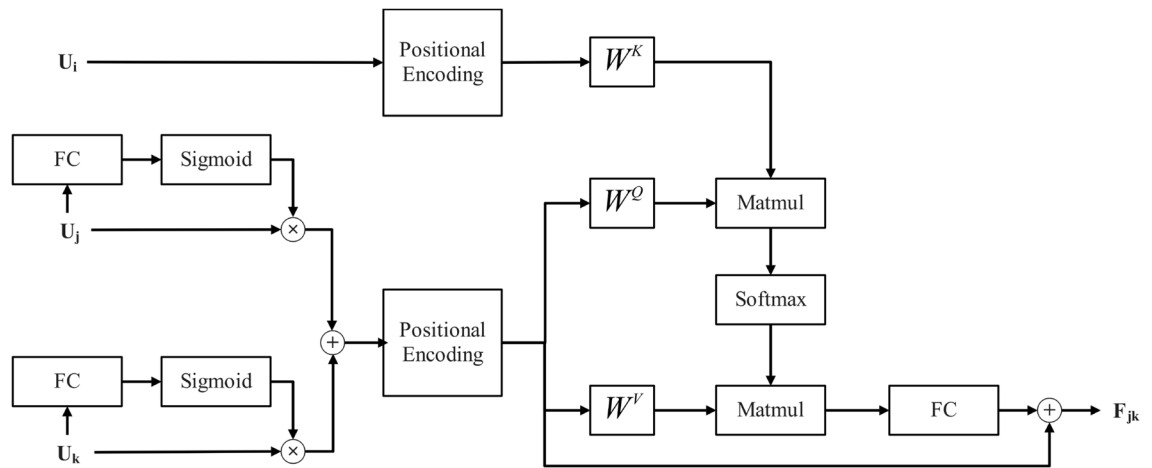


Fig. 3. The proposed gated multimodal attention.

The classifier module

The classifier decision module generates the final emotion prediction results by feeding the fused features of the three sets of bimodal features into the emotion classifier. The proposed method employs two decision-making strategies during classification, namely bimodal fusion prediction and trimodal gated fusion prediction. During the training phase, both strategies simultaneously output prediction results to learn label information, while knowledge distillation is used to enhance the representation capability of unimodal features. However, during the inference and testing phase, only the prediction results from the trimodal gated fusion features are used as the final output. The classifier structure follows a “*FC + Softmax*” design to generate a probability distribution. Specifically, for the trimodal gated fusion module, the feature extraction process is formulated as follows:

$$F_{\text{merge}} = \sum_{m \in \{at, av, vt\}} \text{softmax}(P) * F_m \quad (7)$$

where P represents the trimodal fused features and can be expressed as follows:

$$P = \text{concat}(FC(F_{at}), FC(F_{av}), FC(F_{vt})) \quad (8)$$

The proposed loss function

Since soft labels contain rich implicit knowledge that can be used for training supervision⁴⁵, this study introduces a self-distillation mechanism to transfer knowledge from both hard labels (i.e., emotion labels) and soft labels (i.e., teacher network outputs) to each modality, guiding the model to learn more expressive modality representations. In this approach, the output of the trimodal fused features serves as the teacher model's output, while the three sets of bimodal fusion features act as the student model outputs to learn improved modality representations. The corresponding loss function is formulated as follows:

$$L = \gamma_1 L_{\text{sup-t}} + \gamma_2 L_{\text{sup-s}} + \gamma_3 L_{\text{dist}} \quad (9)$$

where $L_{\text{sup-t}}$ represents the loss of the teacher model output relative to the hard labels, $L_{\text{sup-s}}$ denotes the loss of the three bimodal student models relative to the hard labels, and L_{dist} corresponds to the loss of the student model outputs relative to the teacher model outputs. All three loss functions are implemented using cross-entropy loss. Similar to previous works, this paper employs a temperature coefficient τ to generate soft labels $\hat{Y}_t^\tau$, which can be expressed as follows:

$$\hat{Y}_t^\tau = \text{Softmax}\left(\frac{P}{\tau}\right) \quad (10)$$

The hyperparameters γ_1 , γ_2 and γ_3 are used to balance the contributions of the three types of losses. In this study, all three parameters are set to 1 in the experiments.

Results and discussion

Experimental datasets and evaluation metrics

This study employs the IEMOCAP and MELD datasets to evaluate the proposed model. The IEMOCAP dataset consists of speech data from dyadic conversations, collected from 10 volunteers, comprising a total of 153 dialogues and 7433 utterances. The dataset is divided into five session groups, with the first four groups used for training and the last group reserved for testing. It includes six emotion categories, namely happy, sad, neutral, angry, excited, and frustrated.

Methods	Happy		Sad		Neutral		Angry		Excited		Frustrated		ACC	w-F1
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1		
Dialogue RNN ³⁷	57.6	57.6	78.0	80.3	75.5	70.6	68.2	65.0	73.9	76.0	59.1	62.4	69.4	68.4
Dialogue TRM ⁴⁰	72.2	62.8	85.7	83.3	69.3	68.1	79.4	66.7	67.2	75.0	57.2	63.3	69.9	69.9
MMGCN ⁴¹	50.0	56.3	78.8	81.4	71.4	67.6	68.2	66.3	75.9	76.8	65.1	64.9	69.6	69.6
MMDFN ⁴²	57.6	52.9	84.5	86.1	76.0	71.7	70.6	65.0	73.2	75.3	55.9	62.2	69.9	69.9
SDT ⁴⁵	72.2	63.8	80.0	82.7	71.4	73.4	72.4	69.9	77.3	80.8	70.9	69.7	73.8	74.1
AR-IIGCN ⁵¹	71.9	69.9	74.6	81.6	67.3	63.8	73.8	68.4	82.7	79.2	60.4	63.9	70.5	70.4
MERC-PLTAF ⁵²	75.6	75.3	74.2	80.0	71.1	71.4	75.9	74.7	59.7	54.7	73.5	72.2	72.7	71.4
SACCMAS ⁵³	30.6	38.6	80.0	86.5	70.3	64.9	60.6	64.4	80.9	74.4	63.0	63.0	67.5	65.3
Ours	72.2	66.2	81.6	82.8	76.6	75.0	74.7	69.8	77.9	80.9	64.8	68.1	74.3	74.3

Table 1. Result comparison on IEMOCAP dataset. Significant values are in bold.

Methods	Neutral		Surprise		Fear		Sadness		Joy		Disgust		Anger		ACC	w-F1
	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1		
Dialogue RNN ³⁷	85.1	79.6	54.1	56.7	10.0	12.7	29.8	38.6	62.9	63.8	22.1	27.3	53.6	53.2	66.7	65.3
Dialogue TRM ⁴⁰	83.4	79.5	54.5	57.1	24.0	27.9	33.2	41.0	60.5	62.8	22.1	28.0	58.3	54.0	66.7	65.8
MMGCN ⁴¹	81.5	79.2	58.4	57.8	8.0	13.8	31.7	39.4	69.9	63.4	20.6	24.6	52.2	53.5	66.4	65.2
MMDFN ⁴²	83.5	79.7	63.4	58.2	32.0	26.7	26.4	35.7	63.7	64.9	19.1	24.8	49.3	52.2	66.6	65.5
SDT ⁴⁵	82.6	80.1	61.2	59.3	12.0	16.0	36.5	42.8	61.9	64.1	25.0	30.1	56.5	53.5	67.1	66.4
AR-IIGCN ⁵¹	81.1	81.2	56.2	57.2	6.90	5.1	47.4	37.3	65.9	65.9	2.94	2.94	42.3	45.2	64.1	64.0
MERC-PLTAF ⁵²	82.9	80.5	59.1	59.1	24.0	26.9	55.3	46.4	63.4	77.1	17.6	26.1	50.3	53.9	68.0	52.8
SACCMAS ⁵³	80.6	79.1	47.3	48.0	14.0	19.8	3.8	6.7	54.2	56.5	3.1	5.3	53.4	46.1	62.4	37.4
Ours	83.9	80.0	61.9	58.7	10.0	14.5	32.2	41.7	63.4	64.6	22.1	29.1	55.1	53.7	67.4	66.2

Table 2. Result comparison on MELD dataset. Significant values are in bold.

The MELD dataset consists of multiple conversational segments collected from the TV series Friends, comprising a total of 1433 dialogues and 13,708 utterances. This dataset contains seven emotion categories, including neutral, surprise, fear, sadness, joy, disgust, and anger.

This paper employs accuracy and weighted average F1-score as the primary evaluation metrics to assess the overall performance of different methods on the emotion recognition task. Additionally, the recognition performance for each individual emotion category is evaluated separately to ensure a more comprehensive comparison and analysis of the model’s effectiveness across different emotion recognition tasks.

Result comparison with state-of-the-art methods

The proposed method is compared with state-of-the-art algorithms on both the IEMOCAP and MELD datasets. As shown in Tables 1 and 2, ACC, F1, and w-F1 represent accuracy, F1-score, and weighted F1-score, respectively. It can be observed that the proposed method outperforms state-of-the-art algorithms in terms of overall accuracy and weighted F1-score. The Transformer architecture is more effective at capturing long-range dependencies in sequential data, which explains its superior performance in accuracy and weighted F1-score compared to Dialogue RNN, which primarily relies on RNN for feature extraction. Although graph neural networks (GNNs) are well-suited for modeling inter-modal relationships, they do not provide a significant advantage in intra-modal feature extraction. As a result, MMGCN and MMDFN achieve lower overall accuracy and weighted F1-score compared to Transformer-based approaches.

Dialogue TRM, SDT, and the proposed method all utilize Transformer as the primary feature extraction architecture; however, they exhibit significant differences in extracting inter-modal dependencies. Dialogue TRM primarily integrates multimodal features through a gating mechanism at the decision stage. SDT extracts intra-modal dependency features only at the mid-stage of feature extraction. In contrast, the proposed method not only captures intra-modal dependencies but also integrates the dependency relationships between unimodal and multimodal features, enabling the extraction of more expressive and informative representations.

The confusion matrices of the proposed method on the IEMOCAP and MELD datasets are shown in Figs. 4 and 5, where the horizontal axis represents the predicted classes, and the vertical axis represents the ground truth labels. In the IEMOCAP dataset, there are potential misclassifications between negative and positive emotion labels, such as “happy” and “excited”, or “sad” and “frustrated”. This phenomenon is similar to human misjudgment in emotion perception. Future work will focus on improving fine-grained feature extraction to enhance the classification accuracy of similar emotions.

We performed comprehensive evaluations on the IEMOCAP and MELD datasets against several recently published methods, including Adversarial Representation with Intra-Modal and Inter-Modal Graph Contrastive for Multimodal Emotion Recognition (AR-IIGCN)⁵¹, Multi-modal Emotion Recognition in Conversation based

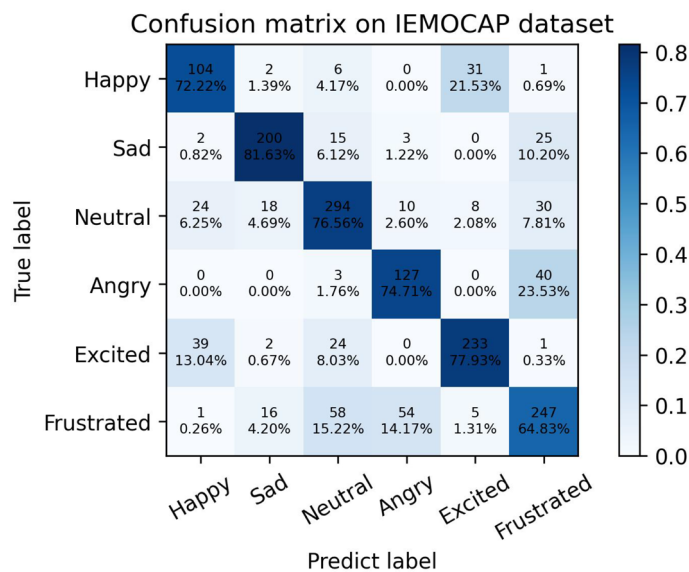


Fig. 4. Confusion matrix on IEMOCAP dataset.

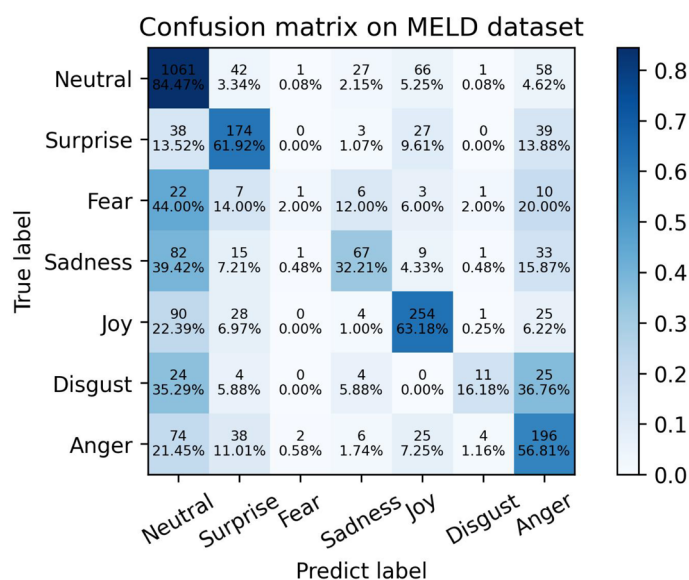


Fig. 5. Confusion matrix on MELD dataset.

on Prompt Learning with Text-audio Fusion Features (MERC-PLTAF)⁵², and Speaker-Aware Cognitive network with Cross-Modal Attention (SACCMA)⁵³. As summarized in Table 1 and Table 2, our model demonstrates consistently strong or leading results across both benchmarks. In particular, it achieves the highest weighted F1 on IEMOCAP (74.3%), highlighting its capability to accurately classify both high-frequency and minority emotion categories. For MELD, our method attains the top weighted F1 score (66.2%) and maintains robust recall and F1 values across multiple emotions, notably Joy and Anger, where competing approaches tend to have a more pronounced decline in performance. These findings indicate that the proposed cross-modality gating framework supports more stable and balanced emotion recognition across diverse settings.

In the MELD dataset, a notable misclassification issue arises, where emotions are often misclassified as “neutral.” This is primarily due to the significant class imbalance in MELD, where the “neutral” class contains a disproportionately high number of samples. This imbalance impacts not only our method but also other models, as shown in Table 2. Despite our model achieving the highest accuracy on MELD, it ranks second in terms of weighted F1-score. This discrepancy highlights the challenges posed by the dominance of neutral emotions in conversational data, which leads to biased performance, especially for less represented emotional categories. While our model achieves state-of-the-art results on the IEMOCAP dataset, addressing this class imbalance remains a crucial focus for future research in emotion recognition, particularly for datasets like MELD.

Methods	ACC	w-F1
baseline	69.6	69.5
+ Cross-Modality Gating	70.5	70.7
+ L_{sup-s}	72.2	72.2
+ L_{dist}	74.3	74.3

Table 3. Ablation studies.

True	Pred					
	Happy	Sad	Neutral	Angry	Excited	Frustrated
Happy	66.4 72.2	2.1 1.4	5.5 4.2	0 0	24.7 21.5	1.4 0.7
Sad	1.2 0.8	75.8 81.6	6.9 6.1	2.0 1.2	0.8 0	13.3 10.2
Neutral	7.2 6.3	5.7 4.7	69.3 76.6	3.9 1.8	3.9 2.1	10.1 7.8
Angry	0 0	0.6 0	3.5 1.8	69.4 74.7	0 0	26.6 23.5
Excited	14.9 13.0	1.0 0.7	9.6 8.0	0 0	70.0 77.9	4.6 1.7
Frustrated	0.5 0.3	4.7 4.2	16.8 15.2	15.7 14.2	1.8 1.3	60.4 64.8

Table 4. Comparison of confusion matrices on IEMOCAP. Each cell shows the “Baseline | Ours” results.

Ablation studies

To further validate the effectiveness of each proposed sub-method, an ablation study was conducted, with the results presented in Table 3. The ablation experiments were performed on the IEMOCAP dataset using an incremental approach, where improvements were added step by step. The “baseline” refers to the network structure without the proposed gating mechanism, where only fused features were used for training, optimizing L_{sup-t} . After incorporating the proposed cross-modal gated mechanism, the model’s accuracy improved by approximately 1.0%. Furthermore, after introducing two additional loss functions, accuracy increased by 1.7% and 1.1%, respectively.

After incorporating the cross-modal gated module, the dependency between modalities and the representation of fused features were further enhanced. However, the feature representation capability of individual modalities remained unoptimized. With the introduction of the L_{sup-s} and L_{dist} loss functions, unimodal feature representations were effectively refined, ultimately leading to a comprehensive improvement in both overall accuracy and F1-score.

We compare the confusion matrix of the baseline and our method in Table 4. We notice that our model shows improved classification accuracy in emotion categories of Neutral, Frustrated, and Sadness, where the baseline model had more misclassifications. Specifically, Frustrated and Neutral emotions are now better separated, while Happy and Excited classifications show fewer errors. We deduce this is due to the introduction of the Gated Multimodal Attention mechanism, which allows the model to dynamically adjust the influence of each modality for different emotions, resulting in improved clearer separation between similar categories. Additionally, the Cross-Modality Distillation Loss improves the consistency across modalities, helping the model reduce misclassifications, especially in more complex emotions.

To better understand how the cross-modality gating mechanism behaves under different emotional conditions, we examined the average modality weights across various emotion categories. As shown in Fig. 6, the weights remain largely balanced, clustering around 0.33. This is a natural outcome of the softmax operation, suggesting that the model generally treats all three modalities with similar importance during the fusion.

We notice that some small but meaningful differences are evident. First of all, text tends to carry more weight in emotions of Neutral, Surprise, and Joy, hinting that verbal cues are more informative in these positive or neutral states. Conversely, Fear, Sadness, and Disgust place slightly more emphasis on audio, likely due to the richness of vocal signals in expressing negative emotions. Visual cues remain relatively steady across categories, though there is a slight uptick in Fear and Disgust, which may be tied to more distinctive facial expressions. Overall, while the gating mechanism does not strongly prioritize any single modality, it does show a modest capacity to adjust modality contributions in response to emotional context.

To assess the impact of different modalities, we compare unimodal, bimodal, and multimodal configurations in Table 5. The text modality outperforms both audio and visual modalities, indicating its significant role in emotion recognition. This advantage may primarily stem from the inherent clarity and lower uncertainty of textual cues, whereas visual cues often introduce higher levels of ambiguity. Although audio modality offers less uncertainty than visual information, it still fails to match the precision and reliability offered by textual data in this context. This is consistent with prior studies. Additionally, combining any two modalities improves performance over unimodal results, with text and audio or text and visual fusion yielding better results than audio and visual fusion alone. Finally, the best performance is achieved when all three modalities (text, audio, and visual) are used together, validating the importance of integrating multimodal information for effective emotion recognition. This demonstrates that combining verbal, vocal, and visual cues is essential for improved emotion classification.

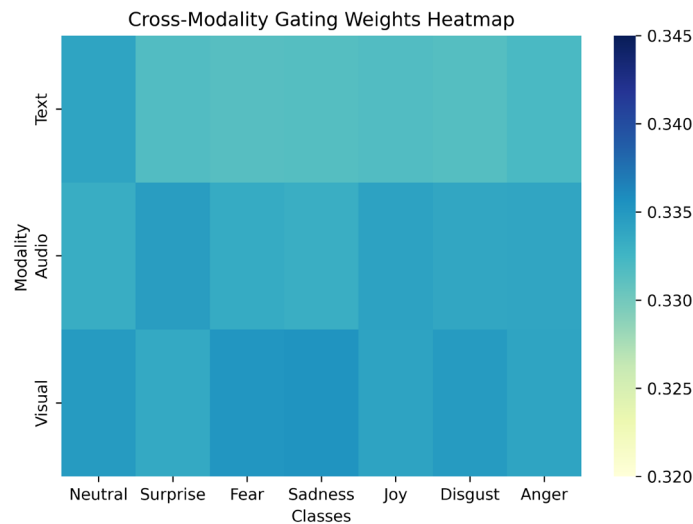


Fig. 6. Cross-modality gating weights averaged by emotion class. While modality weights are generally balanced, text is slightly favored in emotions of Neutral, Surprise, and Joy, and audio in Fear, Sadness, and Disgust. Visual weights remain stable with minor increases in Fear and Disgust.

Modality	IEMOCAP		MELD	
	ACC	w-F1	ACC	w-F1
Text	65.4	65.6	65.9	64.5
Audio	59.8	59.9	48.7	41.2
Visual	42.5	43.0	48.8	31.7
Text + Audio	72.1	72.8	66.8	65.8
Text + Visual	68.3	67.1	66.2	65.9
Audio + Visual	64.0	65.3	49.2	42.2
Text + Audio + Visual	74.3	74.3	67.4	66.2

Table 5. Performance comparison of different modality combinations.

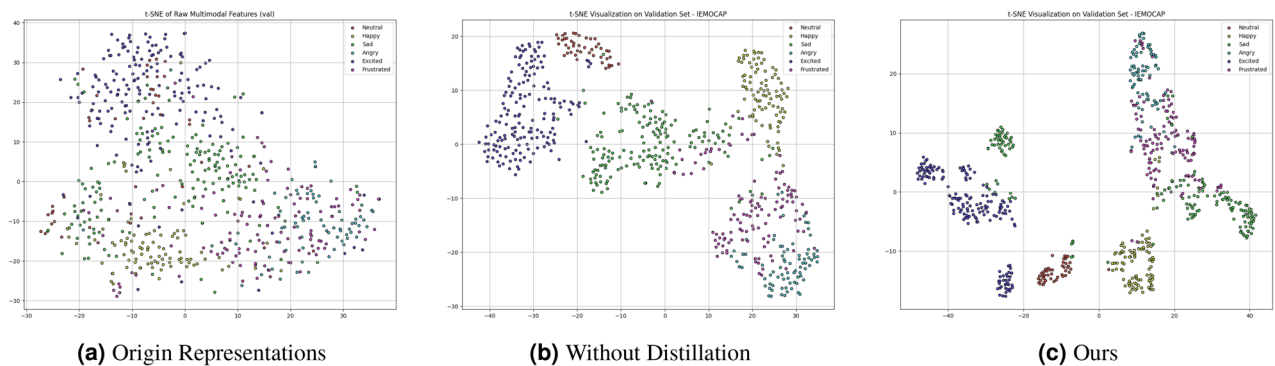


Fig. 7. The t-SNE visualization of the multimodal representations on the IEMOCAP dataset.

We extract multimodal representations for each utterance on IEMOCAP using our proposed transformer-based model, with or without self-distillation. Additionally, unimodal representations are pre-extracted and concatenated to form the original multimodal representations. These representations are then projected onto two dimensions using the t-SNE algorithm.

Figure 7 presents the t-SNE visualizations of these multimodal representations. As seen in Fig. 7a, the original representations show a scattered distribution across emotion categories, with considerable overlap between similar emotions. In Fig. 7b, even without self-distillation, the model slightly improves the separation between emotions, though categories like “Angry” and “Frustrated” remain mixed. Finally, Fig. 7c shows that

incorporating self-distillation results in better separation, with more clearly defined clusters for each emotion. This indicates that self-distillation helps the model generate more distinct multimodal representations, leading to clearer emotion categorization.

Conclusion

This paper proposes a multimodal emotion recognition method based on the Transformer architecture, achieving higher emotion recognition accuracy through the effective fusion of visual, textual, and acoustic features. To enhance feature representation, we design a cross-modal guided fusion attention mechanism, which extracts informative relationships between unimodal features and fused bimodal features. Additionally, a cross-modal distillation loss function is introduced, incorporating a dual-supervision mechanism with both teacher and student models. This approach reduces redundant information across modalities and enhances the model's ability to represent diverse modal features. Experimental results demonstrate that the proposed method outperforms traditional approaches, in terms of accuracy, on both the IEMOCAP and MELD datasets, fully validating its effectiveness in multimodal emotion recognition tasks.

Future research can be further explored in several key directions. First, investigating more effective ways to optimize the feature representation of individual modalities to enhance the accuracy and generalization ability of multimodal fusion-based emotion recognition. Second, addressing the issue of imbalanced emotion labels in conversational emotion recognition, particularly the overrepresentation of neutral emotion samples, which leads to uneven data distribution. Future work will focus on developing balanced training strategies and incremental data processing methods based on adversarial learning to improve the recognition performance of underrepresented emotion categories. These advancements will further enhance the reliability and applicability of multimodal emotion recognition models in real-world scenarios. We also plan to further improve the model by incorporating mechanisms that better capture dialogue dynamics, such as speaker turns, role-specific behaviors, and context across utterances. Specifically, we aim to explore Transformer-based models to better handle inter-utterance dependencies and improve contextual understanding. By investigating ways to more effectively model dialogue flow, this will help enhance the model's performance in real-world conversational scenarios.

Data availability

The IEMOCAP and MELD datasets can be accessed from: IEMOCAP: <https://sail.usc.edu/iemocap/> MELD: <https://affective-meld.github.io/>.

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Author contributions

S.Z. conceived the experiments, S.Z. and X.Z. conducted the experiment(s), S.Z. and J.R. analyzed the results. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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